RAINER JOSEPH B. TAN

tanrainerjoseph9@gmail.com | 09949325055 | Blk 8 Lot 11,Calambeño ville Phase 1, Brgy KayAnlog, Calamba City, Laguna

PROFESSIONAL SUMMARY

Information Technology student with hands-on experience in developing desktop-based applications and managing databases through academic projects. A motivated and adaptable aspiring Application Developer eager to contribute technical skills, solve problems, and grow in the IT industry.

WORK EXPERIENCE

Special Program for Employment of Students (SPES) 2023-2026

Seasonal Employment (1 month annually)

▪ Conducted face-to-face interviews with other people.

▪ Collected and recorded applicant information during the interview process.

PROJECTS

Arduino-Based Smart Cane 2026

Lead Programmer

- Developed the Arduino Uno firmware in C++ by programming five HC-SR04 ultrasonic sensors to measure obstacle distance on the left, center, and right, resulting in distance-based alerts at 20 cm, 50 cm, and 80 cm.
- Programmed ground and stair detection and an MPU-6050 motion sensor to recognize holes, stairs, and uphill or downhill paths, providing the user with advance warning of hazardous terrain.
- Designed the alert priority logic and the message handshake with the mobile application, ensuring one alert is announced at a time and improving the clarity of voice warnings.
- Integrated an ESP32 Bluetooth bridge using serial communication to send alert codes to the smartphone, resulting in wireless communication between the cane and the application.
- Conducted sensor and wiring tests using the Serial Monitor to identify and fix reading errors, improving detection reliability.

Desktop-Based Kiosk System 2022

Lead Programmer

- Developed a desktop-based kiosk application by planning and implementing its overall system structure, resulting in a functional kiosk system.
- Designed the user interface and implemented core application features, providing users with an organized and accessible interface.
- Managed the application database by organizing and storing system records, ensuring structured data management.
- Conducted system testing and debugging to identify and fix errors, improving application functionality.

Desktop-Based School System 2023

Lead Programmer

- Developed a desktop-based school management system by leading its planning, design, and implementation, resulting in a functional school application.
- Designed the user interface and implemented essential system features, providing an organized way to access school functions.
- Managed and organized school records through the database, supporting efficient information storage.
- Performed system testing and debugging to identify and resolve errors, improving system reliability.

Desktop-Based Ordering System 2024

Lead Programmer

- Developed a desktop-based ordering application by handling its planning, design, and implementation, resulting in a functional ordering system.
- Created the user interface and implemented ordering features, providing an organized ordering process.
- Managed order records through the database, ensuring structured storage of transaction information.
- Conducted application testing and debugging to identify and fix errors, improving system reliability.

EDUCATION

2022-2026 Bachelor Of Science In Information And Technology

PWU-CDCEC Brgy Uno, Calamba City

2016-2022 Grade 7-12

Punta Integrated School Brgy Punta, Calamba City

SKILLS & AWARDS

▪**Skills**: VB.NET; MySQL; Database Management; Visual Studio; Debugging; Canva; PowerPoint; Word; Arduino (C/C++)

▪ **Awards**: Service Award